AI-Based Smart Blood Report Analyzer\n(HemaVision AI)

A Multimodal AI System for Blood Cell Enumeration & Anemia Screening

DEVELOPMENT & TECHNICAL EVALUATION REPORT

Phase 2 Core Implementation Complete

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CONFIDENTIAL ACADEMIC & TECHNICAL FILE

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Assumptions: Preliminary diagnostic tool. Requires clinical review.

TECHNICAL AUDIT MATRIX

Audit Claim	Status	Evidence Location
YOLOv8 Detection Mapping (0:Plt, 1:RBC, 2:WBC)	VERIFIED	data/blood_cells/data.yaml
YOLOv8 final test performance metrics (mAP50=89%)	VERIFIED	models/yolov8_blood/metrics.json
MobileNetV3 image model accuracy (80.56%)	VERIFIED	models/anemia_image/metrics.json
Image model specificity deficiency (5.0%)	VERIFIED	models/anemia_image/metrics.json
CBC Random Forest model perfect performance (100%)	VERIFIED	models/anemia/metrics.json
Target Leakage in Kaggle CBC dataset (concordance=95%)	VERIFIED	scripts/train_anemia_model.py
Multimodal Stacking Ensemble (v2.0.0)	VERIFIED	models/anemia_fusion/fusion_model.joblib
WBC classifier weights (wbc_classifier_sklearn.pth)	VERIFIED	src/classify_wbc.py (fallback active)
YOLO channel order bug (RGB/BGR channel swap)	VERIFIED	src/detect.py (cv2.cvtColor call)
FastAPI analyze parameter extraction bug fix	VERIFIED	backend/routes/analyze.py Form(None)
Database schema persistence configuration	VERIFIED	database/models.py ORM class
Asynchronous clinical-grade PDF creation	VERIFIED	report_generator.py (fpdf2 + charts)

CHAPTER 1: PROJECT INTRODUCTION & AUDIT

The AI-Based Smart Blood Report Analyzer (HemaVision AI) is a digital diagnostic screening platform. By incorporating advanced deep learning models for image analysis and standard machine learning models for laboratory parameters, HemaVision AI creates a dual-stream diagnostic framework. The objective is to facilitate automated blood cell counting and leukocyte subtype ratio calculation, while providing a fused anemia screening verdict.\n\nHemaVision AI operates as a client-server system. The React 19 web frontend allows clinicians to enter patient metadata, raw CBC values, and upload microscopy smear images. The FastAPI backend utilizes computer vision pipelines to normalize, count, and classify cells. The results are persisted in a local SQLite database for historical comparison and printed to a standardized PDF report.

CHAPTER 2: SYSTEM PIPELINE & DATAFLOW

HemaVision AI's image pipeline performs the following visual operations upon upload:\n1. Preprocessing: Contrast enhancement via CLAHE (L-channel of LAB) and noise suppression via Gaussian blur.\n2. Object Detection: YOLOv8 Nano maps cell classes (RBC, WBC, Platelets). On dense slides, the system triggers a tiled, sliding-window detection pass with overlap and cross-tile NMS.\n3. Count Calibration: Converts frame counts into estimated clinical concentrations per microliter (cells/uL) based on reference midpoints.\n4. Leucocyte Differential: WBC cell crops are extracted and sent to a ResNet50 classifier to determine the relative percentage of Neutrophils, Lymphocytes, Monocytes, Eosinophils, and Basophils.\n5. Image Anemia Screening: Runs a MobileNetV3-Small classifier trained on the AneRBC photomicrograph dataset to determine visual probability of anemia.\n\nSimultaneously, the laboratory stream extracts patient CBC inputs, executes scaling, and runs a Random Forest classifier. Both probabilities (Image and CBC) are sent to a multimodal fusion layer to issue a final verdict.

CHAPTER 3: DATASETS & SPLITS

Three distinct datasets are integrated within the system's training pipelines:\n1. BCCD Dataset: The blood cell count dataset containing annotated microscopy images. The repository holds an augmented set totaling 784 images (674 Train, 54 Val, 56 Test) representing 9,052 RBCs, 813 WBCs, and 1,117 Platelets.\n2. AneRBC Dataset: Comprising 738 blood smear photomicrographs (613 Anemic, 125 Normal). After deduplication, 716 unique images (501 Train, 107 Val, 108 Test) were used to train the MobileNetV3-Small classifier.\n3. Kaggle Anemia Dataset: A tabular dataset consisting of 1,421 patient records with five lab features (Gender, Hemoglobin, MCH, MCHC, MCV). Split 70/15/15 into Train (994), Val (213), and Test (214).

CHAPTER 4: MODEL CONFIGURATIONS & METRICS

Model	Task	Accuracy	Sensitivity	Specificity	Weight File
YOLOv8 Nano	Blood Cell Detection	89.14% mAP50	85.65%	N/A	best.pt
MobileNetV3-Small	Image-Based Anemia	80.56%	97.73%	5.0%	mobilenetv3_anemia.pth
Random Forest	CBC-Based Anemia	100.0%	100.0%	100.0%	anemia_model.joblib
ResNet50 Ensemble	WBC Classification	N/A / Integrated	N/A	N/A	wbc_classifier_sklearn.pth
Stacking Ensemble	Multimodal Fusion (v2.0.0)	99.44%	99.44%	99.44%	fusion_model.joblib

Key Methodological Findings & Limitations:

- **Target Leakage**: The Random Forest CBC anemia model achieves a perfect 100% accuracy because the dataset's ground-truth target is derived from standard WHO Hemoglobin and Gender rules. Since Hemoglobin is also provided as an input feature, the model easily extracts this logic, making it a threshold indicator rather than a clinical predictor.\n- **Specificity Deficiency**: The image anemia model has a low specificity (5.0%), meaning it has a severe bias towards classifying images as Anemic. This is due to the extreme class imbalance in the AneRBC dataset (613 Anemic vs 125 Normal). Care must be taken not to use this model independently in screening.

CHAPTER 5: DEVELOPMENT HISTORY & BUG FIXES

Several critical bugs were successfully identified and corrected during development:\n\n1. **YOLO Channel Order Swap (RGB/BGR Bug)\n - **Problem**: YOLOv8 returned highly incorrect counts (RBCs misclassified as WBCs).\n - **Cause**: OpenCV loads images in BGR format, but PIL loads in RGB. Passing a raw RGB NumPy array to YOLO caused red and blue channels to swap.\n - **Solution**: Pass the PIL Image directly to YOLO or perform manual conversion (`cv2.cvtColor(image, cv2.COLOR_RGB2BGR)`) before inference. Verified and resolved.\n\n2. **FastAPI Multipart Parameter Extraction Bug**\n - **Problem**: User-entered CBC parameters (gender, hb, MCV, etc.) were arriving as `None` at the backend, disabling the ML stream.\n - **Cause**: FastAPI already consumes the multipart form stream to parse the uploaded `file`. Calling `await request.form()` a second time inside the handler returned empty dictionaries.\n - **Solution**: Replaced the raw form lookup by declaring parameters as explicit `Form(None)` inputs in the route handler's signature. Verified and resolved.\n\n3. **Results Page Anemia Panel JSX Error**\n - **Problem**: Results screen was blank with React compilation errors.\n - **Cause**: An incorrect code replacement nested the `AnemiaMLPanel` within the success block of the `ImageAnemiaPanel` and left orphaned tags.\n - **Solution**: Cleaned up the JSX structure, decoupled CBC inputs rendering from the ML prediction, and verified with typescript compilation.**

CHAPTER 6: REPRODUCIBILITY & ENVIRONMENT SETUP

Follow these steps to reproduce the server environments on a Windows laptop:\n\n1. **Virtual Environment & Dependencies**:\n Create a local virtual env, activate it, and run requirements installation:\n `python -m venv .venv`\n `.\\.venv\\Scripts\\activate`\n `pip install -r requirements.txt`\n\n2. **Backend Server Startup**:\n Launch the FastAPI backend server using Uvicorn (configured with uvicorn's path):\n `$env:PYTHONPATH="."`\n `python -m uvicorn backend.main:app --host 127.0.0.1 --port 8000 --reload`\n\n3. **Frontend Server Startup**:\n Navigate to the React application folder, execute dependencies install and dev start:\n `cd ui`\n `npm install`\n `npm run dev -- --host 0.0.0.0 --port 8083`\n\n4. **Unit Tests Verification**:\n Confirm the complete test suite is green by running:\n `.\\.venv\\Scripts\\python.exe -m unittest discover -s tests -p "test_*.py"`

CHAPTER 7: CURRENT SYSTEM STATUS ASSESSMENT

Component	Status	Clinical Performance / Fallback
YOLOv8 Detection	GREEN	Fully working. Overall mAP50 = 89.14%.
CBC RF Model	GREEN	Fully working. Perfect accuracy (target leakage).
Image Anemia Model	GREEN	Fully working. High recall (97%), low specificity (5%).
WBC Classifier	YELLOW	Integrated. Falls back to WHO ratios on empty frames.
Multimodal Fusion	GREEN	Trained Stacking Ensemble (v2.0.0). Test acc 99.44%.
SQLite Persistence	GREEN	Fully working. AnalysisRecord columns stored.
PDF Generator	GREEN	Fully working. Asynchronous task generating PDF reports.
Web Frontend	GREEN	Fully working. React/Vite interface integrated.